Guidance & Feedback Computer-Mediated Reality Systems in the Energy Industry

Deploying Augmented Reality in Nuclear Generating Stations

A Project Proposal for OPG Ignite





Introduction

Computer-Mediated Reality (CMR) will completely revolutionize the way data and information is presented on the factory floor.

Engineers and technicians at electricity utilities are often held to the highest performance, efficiency and safety standards in the energy industry. The nuclear inspection trade is no different.

Nuclear inspection personnel frequently face scenarios with increased radioactive exposure risks, where attention to detail and procedural adherence are equally paramount. Access to vital information, including online dose measurements and instruction manuals, can improve efficiency and safety and reduce safety related incidents by at least 20%. **This project will explore the deployment of Computer-Mediated Reality Systems to provide critical data feedback and procedural guidance to inspection personnel in the nuclear industry.**

The Current State of Nuclear Inspection Technology

Nuclear power plants require a range of technologies across land, sea, and air to ensure safe and reliable operation. The bulk of nuclear inspections involve specialized, non-destructive tooling, harnessing ultrasonic, phased array ultrasonic, eddy current, laser scanning, Light Detection and Ranging (LIDAR), Radio Detection and Ranging (RADAR), and visual inspection technologies. These tools are deployed using delivery machines that include rovers, robotic delivery systems, and drones.

Nuclear Inspection personnel are often required to report to environments with elevated potential risk of exposure to radiation. While the use of personal dosimetry devices help to mitigate these risks by providing dose feedback to technicians, radiation mapping would provide, maintenance technicians, operations crews and radiation protection staff with an understanding of localized radiation zones and escape routes. These improvements would optimize the persons ability to foresee a conservative bias for work in an area of high potential of radiation dose. Current dosimetry devices offer a limited, ineffective user interface which can be easily ignored.

Further, nuclear energy workers rely on continuous access to manual guides: Equipment Inspection & Maintenance procedures, Workplans / Workorders, etc. The process of controlling and executing the work processes that consist the use of plans and/or procedures tends to be devious. These plans – especially work requiring the inspection of equipment on several systems – must be adhered to ensure safe and efficient work. Whether it be turning a harmless valve to maintaining a critical component, CMR will help optimize the ability for nuclear energy workers to communicate important information including procedural adherence, equipment health statuses, and reinforce key decision making skills in-field.



What is Computer-Mediated Reality?

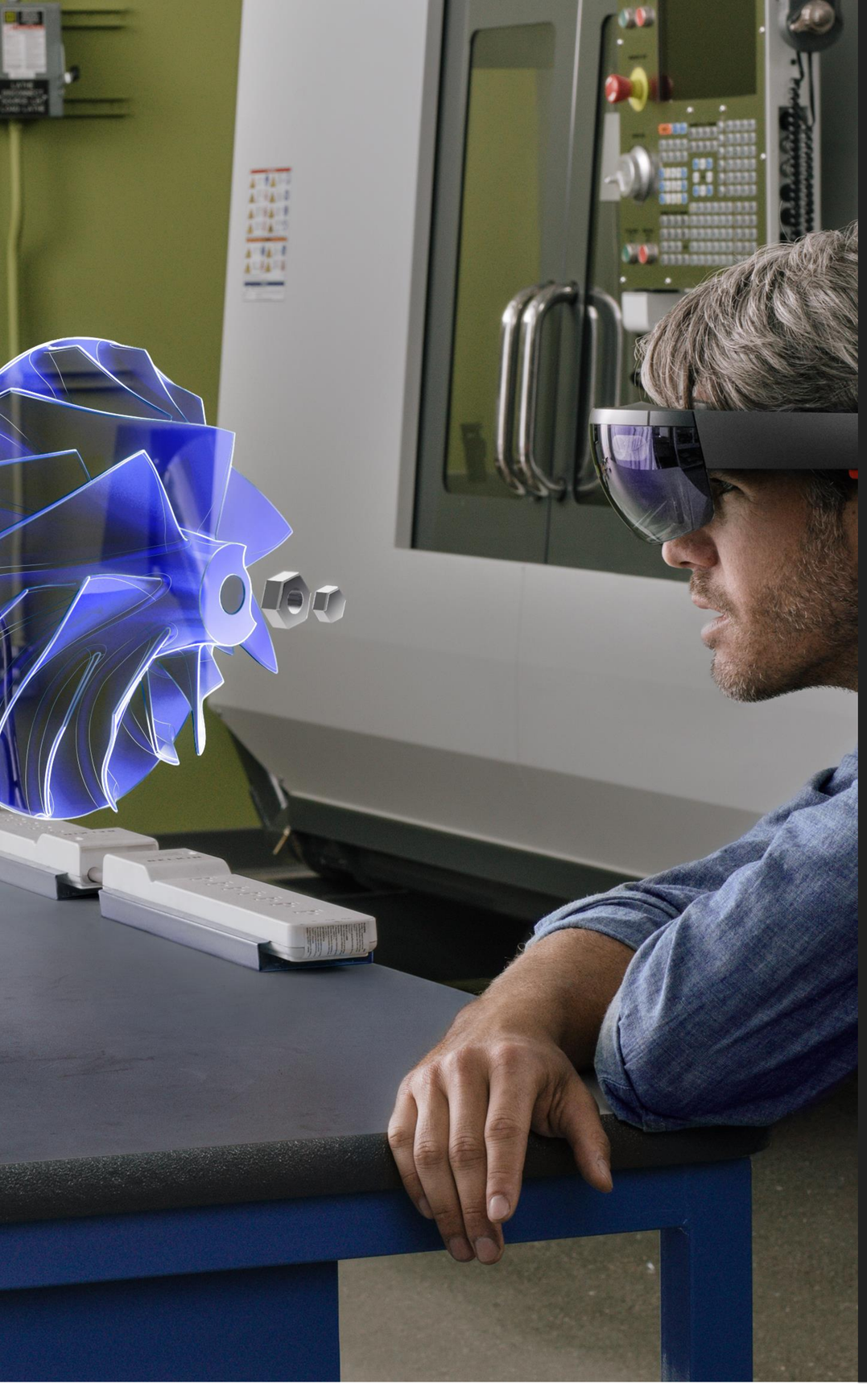
Computer-Mediated Reality (CMR) is the ability to manipulate reality with the aid of a portable computer or mobile device. CMR is a branch of computer vision, an exciting field that explores the use of computers to understand detailed digital media. By combining a real-time acquisition device (a camera or microphone) with a powerful processing computer (a mobile device), intelligent software can perform high-level applications including object identification, video tracking, and reality augmentation.

The past decade has seen the development of several motion tracking devices. Primarily from the video gaming industry, examples include the Nintendo Wii (2006), PlayStation Eye/Move (2007/2010), and Microsoft Kinect (2010). These consumer devices enable gesture recognition through a combination of accelerometers and rate sensors to track motion, and an infrared camera to track position. Together they detect pushing, pulling, rotating and lifting actions.

More recent development in CMR technology has led to the rise of Head Worn Displays (HWD). These systems bring complex feedback data and information to the user's line of sight. HWDs differentiate from Head Mounted Displays (HMD) by how they are worn. HMDs are typically dual, high-resolution displays mounted over the eyes and held onto the head using a strap. Great examples of HMDs include the Google Cardboard (2014), Oculus Rift (2016), HTC Vive (2016) and Microsoft HoloLens (2016). HWDs extend the ergonomics of HMDs in that they can be worn more easily; as easy as putting on glasses. Examples of HWDs are Epson Moverio (2012), Vuzix M100 (2012), and Google Glass (2013).

HWD and HMD devices harness different types of CMRs depending on their use and purpose.





Four Different Computer-Mediated Realities

There are four different types of CMRs; virtual, augmented, mixed, and diminished realities. Each reality provides a unique experience; they solve different problems and can be applied to diverse sections in the nuclear industry.

1

VIRTUAL REALITY

Virtual Reality, or VR, is a completely immersive experience that simulates an entire room within which the user can manipulate interactive, virtual objects.
Devices: HTC Vive, Oculus Rift
Applications: Training

2

AUGMENTED REALITY

Augmented Reality, or AR, superimposes a virtual, interactive object *onto* the user's line of sight. The user often interacts using voice commands and basic gestures.
Devices: Google Glass, Moverio
Applications: Procedures, Radiation Data Feedback

3

MIXED REALITY

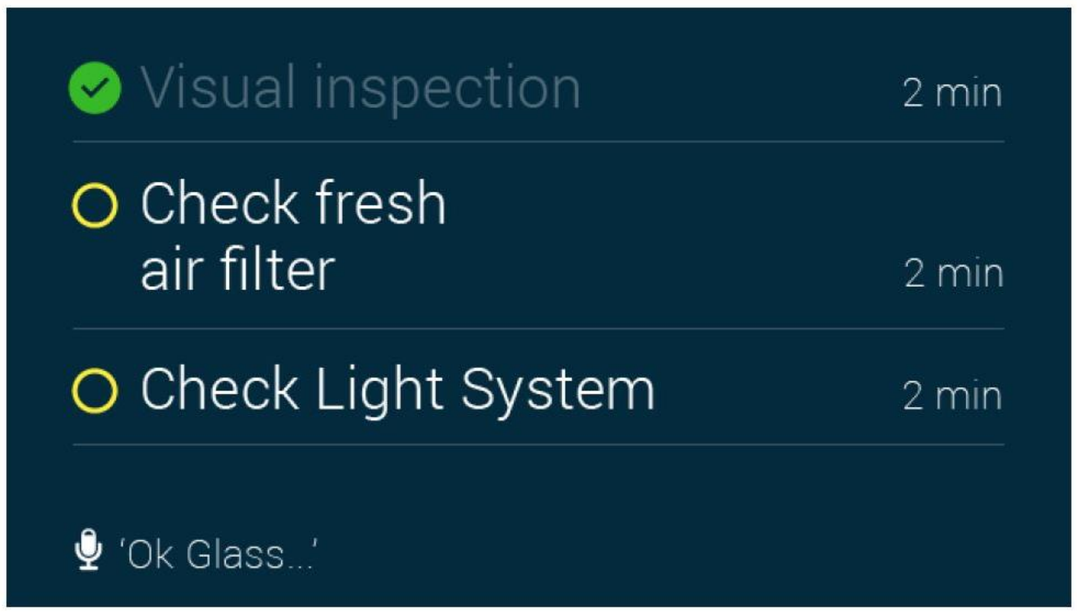
Mixed Reality, or MR, superimposes a virtual, interactive object onto the user's line of sight. The difference between AR and MR is that a user can interact with the object using complex hand gestures.
Devices: Microsoft HoloLens
Applications: GIS, Training

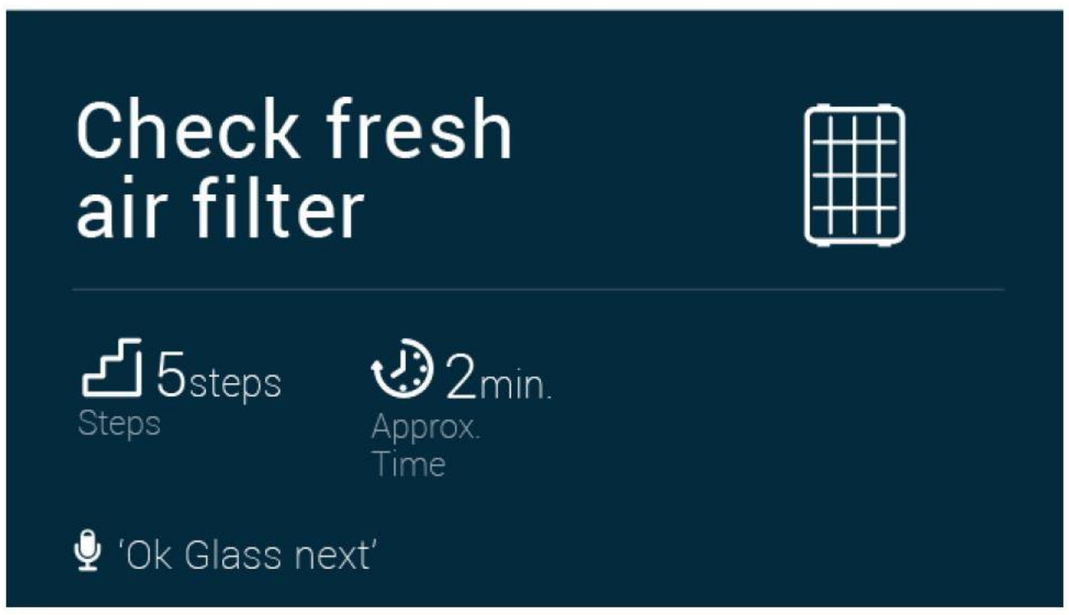
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DIMINISHED REALITY

Diminished Reality, or DR, is an obscure and novel field in CMR. It involves the removal or hiding of physical objects in the virtual world. It's the opposite of Augmented Reality.
Devices: None
Applications: GIS, Valve/Pipe Flow Control

Project Objective







While any of the four CMRs can be applied to the energy industry, this project will focus on developing radiation mapping applications (apps) for AR devices like Google Glass.

These applications will provide a hands-free experience for maintenance technicians, allowing them to easily obtain the following information:

Radiation Dose Feedback & Mobile Escape Guidelines

An app for real-time display of dose rate and total dose from personnel dosimeters. This can interface with another app which provides escape and back-out guidelines in event of high dose/disaster events.

Workflow Guidance

An app that provides visual guidance including checklists, contextual menu and support, and notifications. Using voice control, technicians can progress through steps.

Virtual Collaboration

An app that enables real-time collaboration between field and office crews. Technicians can trigger video conference calls when facing challenging issues requiring support from technical experts.

Project Outline

This project will be conducted in stages and involves a lot of preliminary research on various AR devices and their capabilities, dosimeters and real-time upload, and ergonomic User-Experience/User-Interface (UX/UI) design. Nuclear shielding, personal protective equipment (PPE), and radiation degradation will also be evaluated. The Agile development model will be used to encourage aligning the project's goals with that of the inspection industry.



RESEARCH

Considerable research on AR devices, real-time dosimetry, UX/UI, electronics shielding, and PPE to develop housing for the portable computing device.



BUILD KNOWLEDGE DATABASE

A knowledge base will be developed to store real-time dosimetry data, disaster escape routes, and inspection procedures.



DEVELOP LIDAR MAPS OF TESTING ENVIRONMENT

The LIDAR mesh map will provide an indoor, geolocation-like service. It will provide the X.Y. and Z coordinates for radiation measurements collected by the portable dosimeters.



BUILD DOSIMETRY DISPLAY APP

The dosimetry display app will display the dose rate and total dose data indexed in the Knowledge Base from the dosimeter. Will include warnings for high dose counts.



BUILD PROCEDURE GUIDANCE APP

The procedure guidance app will provide checklists, contextual support and notifications to the technician.



ONGOING: BUILD TABLET APPS

Tablets offer a much larger interface for checklists. They make great tools for situations where hands-free interactivity is not required (lab/desk inspections).

This is an ongoing project.

IT advisory firm Gartner Research projects that advancements in AR technology will bring over $1 billion in savings to field service industries. The augmented reality market is expected to grow to over $120 billion by 2020, and more than 72% of companies, such as Idaho National Laboratories, First Energy, and Ontario Power Generation are looking to AR solutions in their business.

Project Evaluation

To determine the success of the project the desired outcomes need to be clearly defined and stated in a non technical manner. The project has many facets each contributing to the overall outcome and success of the project. The desired outcomes for the project include:



01. REAL-TIME DOSE TRACKING

The Knowledge Database needs to be collecting dose rate and total dose data in real-time



02. REAL-TIME DATA FEEDBACK

The Dosimetry Display app must accurately display real-time dose information with minimal latency



03. CLEAN, DISTRACTION-FREE UX

The Procedure Guidance app must correctly and clearly display checklists and notifications in a distraction-free interface.



04. PORTABILITY AND SHIELDING

The CMR processing system must be mobile and easily housed in a protective case, possibly providing radiation shielding as required.

Cost Analysis

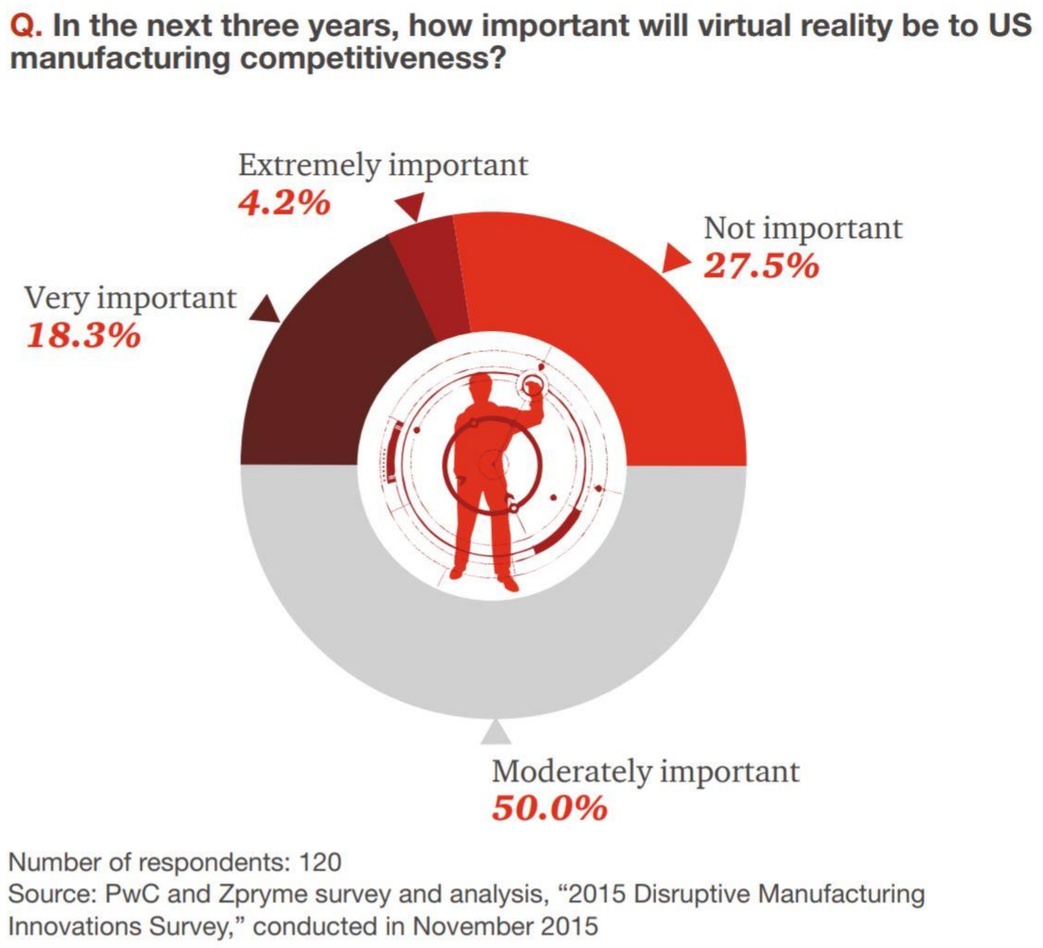
$13,000

Augmented Reality technology is inherently inexpensive, especially compared to Mixed Reality. The most expensive AR device will cost about $3,000 CAD, while the cheapest MR device starts at $5,000 CAD. More expensive does not necessarily mean better features either, the Microsoft HoloLens only has a 2.4MP camera compared to the 5MP 720p camera on the cheaper Google Glass. A cost breakdown of the entire project is as follows:

We project that the entire project can be achieved in under $13,000. This is a small investment compared to the benefits that AR can bring to the inspection industry.

AR DEVICES ($11,000)

For a good understanding of the current AR capability, 4 Google Glass ($8500) + RealWear HMT-1 ($2,000) is required. This will be used to continuously test the radiation feedback application.



DOSIMETERS ($500)

Small portable dosimeters, like the Ultra Radiac™-Plus or DMC-3000 Personal Radiation Monitor, provide real-time data logging. They are inexpensive and reliable.

LIDAR ($400)

A Light Detection and Ranging (LIDAR) camera will generate high quality mesh maps of the testing environment.

KNOWLEDGE SERVER ($1000)

To store real-time data from the portable dosimeters, procedural guidance checklists and notifications, and interface with the AR device. Will be an portable Intel NUC device ($1000).



Thank you for your consideration.